

COURSE: CHE 312: Transport Phenomena II

Spring Semester 2020

Department of Chemical Engineering

University of Illinois at Chicago

Midsemester Exam #1

Only one sheet of paper 5 x 7 is allowed

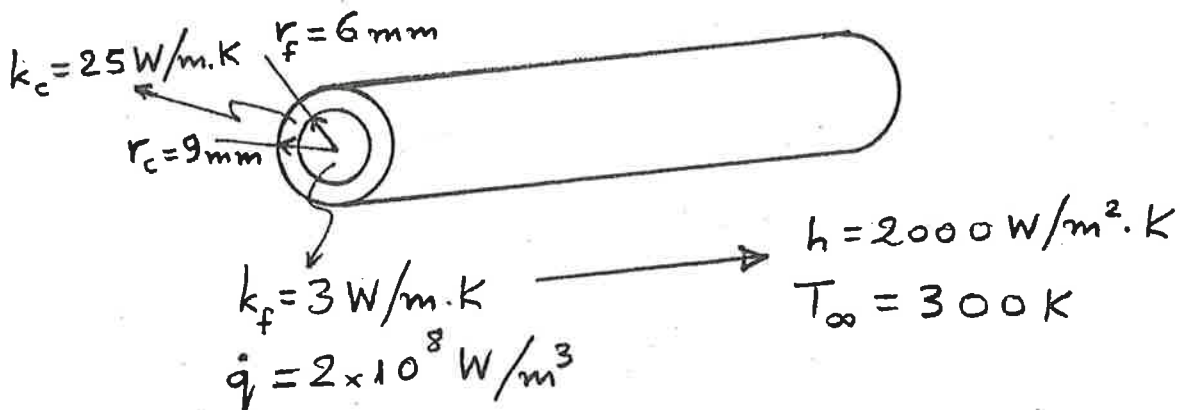
Duration: 50 minutes --- February 10, 2020

Both Problems must be done

Problem 1: (50 points)

A nuclear reactor fuel element consists of a solid cylindrical pin of radius r_f and thermal conductivity k_f . The fuel pin is in good contact with a cladding material of outer radius r_c and thermal conductivity k_c . Consider steady-state conditions for which uniform heat generation occurs within the fuel at a volumetric rate $\dot{q}$ and the outer surface of the cladding is exposed to a coolant that is characterized by a temperature T_∞ and a convection coefficient h .

- (5+5+5+5+5) (50)
- Obtain equations for the temperature distributions $T_f(r)$ and $T_c(r)$ in the fuel and cladding, respectively. Express your results exclusively in terms of the foregoing variables.
 - Consider a uranium oxide fuel pin for which $k_f = 3 \text{ W/m}\cdot\text{K}$ and $r_f = 6 \text{ mm}$ and cladding for which $k_c = 25 \text{ W/m}\cdot\text{K}$ and $r_c = 9 \text{ mm}$. If $\dot{q} = 2 \times 10^8 \text{ W/m}^3$, $h = 2000 \text{ W/m}^2\cdot\text{K}$, and $T_\infty = 300 \text{ K}$, what is the maximum temperature in the fuel element?





Assumptions?

$$a) \frac{1}{r} \frac{\partial}{\partial r} \left(k r \frac{dT}{\partial r} \right) + \dot{q} = 0$$

$$\int d(k r \frac{dT}{\partial r}) = \int -q r dr$$

$$\rightarrow k r \frac{dT}{\partial r} = -\frac{1}{2} q r^2 + C_1 \Rightarrow k \frac{dT}{\partial r} = -\frac{1}{2} q r + \frac{C_1}{r}$$

$$\int dT = \int -\frac{1}{2k} q r + \frac{C_1}{r} \cdot \frac{1}{k} dr$$

$$\rightarrow T(r) = -\frac{1}{4k} q r^2 + C_1 \ln|r| \cdot \frac{1}{k} + C_2 \quad \checkmark$$

B.C.

$$r=0 \Rightarrow \frac{dT}{dr} = 0$$

B.C @ $r=r_f$ $T(r)=T_f(r)$ @ $r=r_c$ $T(r)=T_c(r)$

$$\text{Eqn 1} = T_f = -\frac{1}{4k} q r_f^2 + \frac{C_1}{k} \ln|r_f| + C_2$$

$$\text{Eqn 2} = T_c = -\frac{1}{4k} q r_c^2 + \frac{C_1}{k} \ln|r_c| + C_2 \quad \times$$

Subtract

$$T_f - T_c = -\frac{1}{4k} q (r_f^2 - r_c^2) + \frac{C_1}{k} \ln|r_f|$$

$$C_1 = \frac{k(T_f - T_c) + \frac{1}{4} q (r_f^2 - r_c^2)}{\ln|r_f/r_c|} \quad \times$$

Solving for C_2 yields
 $T_c = T_f$ so $C_2 = 0 \quad \times$

$$T_f(r) = -\frac{1}{4k} q r_f^2 + \frac{[(T_f - T_c) + \frac{1}{4} q (r_f^2 - r_c^2)] \ln|r_f|}{\ln|r_f/r_c|}$$

$$T_c(r) = -\frac{1}{4k} q r_c^2 + \frac{[(T_f - T_c) + \frac{1}{4} q (r_f^2 - r_c^2)] \ln|r_c|}{\ln|r_f/r_c|} \quad \times$$

$$T(r) = -\frac{1}{4k} q r^2 + \frac{[(T_f - T_c) + \frac{1}{4} q (r_f^2 - r_c^2)] \ln|r|}{\ln|r_f/r_c|}$$

Steady State

$$\frac{dT_c}{dr_c} = \frac{-\frac{1}{2k} \dot{q} r_c + (T_f - T_c) + \frac{1}{4k} \dot{q} (r_f^2 - r_c^2)}{r_c^2}$$

b.) $q_{conv} = q_c = q_f$

$$h(T_{\infty} - T_c) = -\dot{q}$$

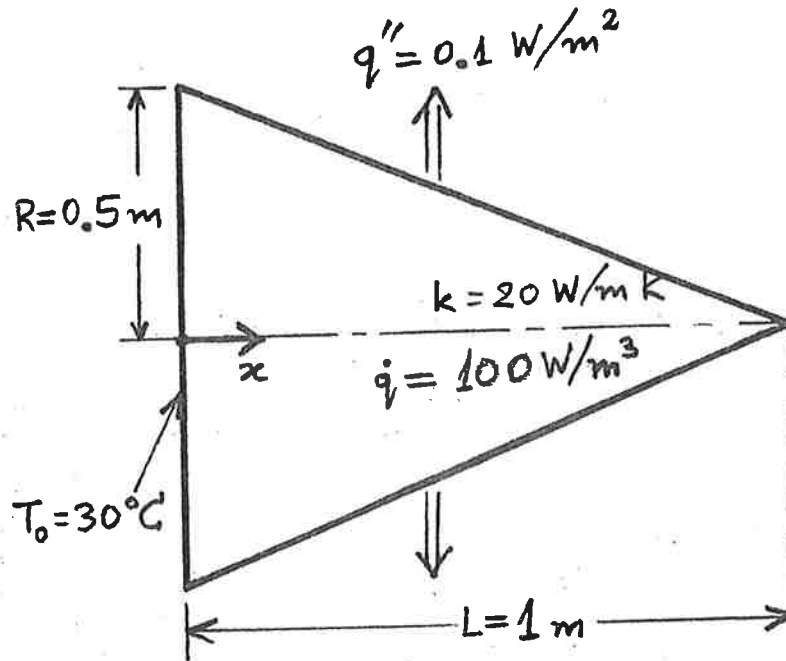
$$h(300 - T_c) = -k \frac{T_{cen}}{r_c} = -k \frac{T_{cen}}{r_f}$$

$$r_f r_c^{-1}$$

$$-r_f r_c^{-2}$$

Problem 2: (50 points)

A solid cone as shown is of circular cross section and the radius of its base is 0.5 m. The cone of length 1 m and thermal conductivity $20 \text{ W/m}\cdot\text{K}$ has uniform volumetric heat generation of 100 W/m^3 . A constant heat flux of 0.1 W/m^2 is leaving the cone through the lateral surface of the cone, while the end surface at $x = 0$ is maintained at $T_0 = 30^\circ\text{C}$. Determine the temperature profile $T(x)$ and the temperature $T(L)$.



$$\frac{d}{dx} \left(k \frac{\partial T}{\partial x} \right) + \dot{q} = 0$$

$$k \frac{dT}{dx} = -\dot{q}x + C_1$$

$$dT = \int -\frac{\dot{q}}{k} x + \frac{C_1}{k} dx$$

$$T(x) = -\frac{\dot{q}}{2k} x^2 + \frac{C_1}{k} x + C_2$$

B.C. @ $x=0$ $T=T_0=30^\circ\text{C}$ & @ $x=L$ $T=T(L)$

$$30 = C_2$$

$$T(L) = -\frac{\dot{q}}{2k} L^2 + \frac{C_1}{k} L + 30$$

Solving for C_1 yields

$$T(L) = T(L) \text{ so } C_1 = 0$$

$$T(x) = -\frac{\dot{q}}{2k} x^2 + 30$$

$$T(L) = \frac{-(100 \text{ W/m}^3)(1 \text{ m}^2)}{(2 \cdot 20 \text{ W/m}^\circ\text{C})} + 30$$

$$T(L) = 27.5^\circ\text{C}$$

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